



SUBMITTED BY: Payal J. Ranva & Yashvi Zazmeriya

ENR. NO : 220451041 & 220451064

FACULTY OF : I.M.C.A

GUIDE BY : Assist. Prof. Uday Shah

Submitted In Partial Fulfillment of,
The Requirements for the Degree of
INTEGRETED MASTER OF COMPUTER APPLICATION

PETNEST

A PROJECT REPORT

Submitted to



Noble University, Junagadh

**For the Practical requirement fulfill in
Integrated Master Of Computer Application
Sem-9**

Prepared by:

Payal J Ranva & Yashvi Zanzmeriya

[Enr.No. – 220451041 & 220451064]

Guided By:

**Mr. Uday Shah
(Assistant Professor)**

Faculty of computer application, Noble University

Noble MCA College, Noble University-Junagadh



Noble University, Junagadh

Certificate

Enrollment No. 220451041

This is to certify that the Project / practical work was satisfactorily carried out and hence submitted this report is the bonafide work of Ms. Payal J. Ranva

Student of Master of Computer Applications, Sem-9 in the Laboratory of Noble University, Junagadh during the academic year 2026-27.

Internal Guide

Principal



Noble University, Junagadh

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Internal Guide

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PREFACE

There is a wide difference between theory and practical. If one has only theoretical background of any subject, one would not succeed in own aim therefore it is necessary for any person to have acceptable practical knowledge of the concerned subject. As I know IMCA is a course based on “Information Technology” and it is totally practical field. With only theoretical knowledge one can’t be succeeded or one can’t be on the peak position.

In the course of IMCA designed by the “Noble University” they have taken full care of these things and designed the course in such a manner with which student can get theoretical and practical both type of knowledge perfectly. According to the rules & regulation of “Web Development”, We have a subject named “PetNest”. In which we have to create a web project of any institute or industry.

As a IMCA student, I have gathered general information about Whole Admission Process. Then We decided to develop the site for that. In this site you can adopt pet very easily. You can also be a volunteer of our NGO.

In this project report We have covered all the information, which is required for the web project of IMCA student.

We have tried as my best to present this project report in such a way that it makes easy to understand the project work.

Acknowledgment

Being students that was really a big task for us to make project. We are very thankful to all those who helped us in preparing this project. We have been fortune for getting the guidance and kind help from our professor.

We are especially thanks to our Dean **Mr. Paresh Vora** and **project guildler** for providing us such a nice platform for the project it is our great pleasure that for providing the chance for the project in HTML, CSS, Bootstrap, Java Script and JSP.

We are extremely thankful and pay our gratitude to our project guide **Assist. Prof. Uday Shah** his valuable guidance and support on completion of this project in its presently.

We extend our gratitude to Faculty of Computer Application, Noble University, Junagadh for giving us this opportunity.

Thanking You.

Payal J. Ranva
Enrollment No. -220451041

Yashvi Zanzmeriya
Enrollment No. -220451041

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1. Project Profile

1.1 Definition

PetNest is a web-based Pet Adoption Management System developed to provide an easy and organized platform for connecting homeless animals with people who are interested in adopting them. The main purpose of the system is to provide information about pets available for adoption and make the adoption process more convenient for users.

Through the PetNest website, users can register and log in, browse available pets, view their details, and submit an adoption request. The system also provides a Volunteer section where users can apply for volunteer opportunities and provide their availability and reason for volunteering.

The project also includes an Admin Panel that helps administrators manage important information such as users, pets, adoption requests, and volunteer applications. Administrators can review applications and update their status according to the organization's requirements.

In addition, PetNest includes a simple AI Assistant that provides quick answers to frequently asked questions related to pet adoption, volunteering, login, events, gallery, contact information, and other basic website-related queries. The system is developed using HTML, CSS, JavaScript, JSP, JDBC, MySQL, and Apache Tomcat.

Overall, PetNest aims to make pet adoption information more accessible, reduce manual record management, and provide a simple digital platform for users and administrators.

1.2 Scope

The scope of the PetNest Pet Adoption Management System covers the major activities required for managing and providing pet adoption services through a web-based platform. The system is designed for users as well as administrators and provides separate functionalities for both.

The main scope of the project includes:

- User Registration and Login: Users can create an account and securely access the features available to registered users.

- **Pet Browsing:** Users can browse the available pets and view information such as pet name, breed, age, description, and image.
- **Pet Details:** Users can select a pet and view its detailed information before submitting an adoption request.
- **Adoption Request:** Registered users can submit an online request for adopting a selected pet along with the required information and reason for adoption.
- **Volunteer Application:** Users can apply for volunteer opportunities by providing their personal details, reason for volunteering, and availability.
- **Admin Management:** The administrator can manage users, pets, adoption requests, and volunteer applications through the Admin Panel.
- **Application Status Management:** Administrators can review adoption and volunteer applications and update their status, such as pending, approved, or rejected.
- **Events and Gallery:** The website provides sections for displaying events and gallery content related to PetNest activities.
- **Success Stories:** The system provides a section for sharing successful adoption stories and creating awareness about pet adoption.
- **About and Contact Information:** Users can learn about the organization, its mission, and contact details.
- **AI Assistant:** A simple AI Assistant is included to provide quick responses to common questions about adoption, volunteering, contact details, login, events, and other website features.
- **Database Management:** MySQL is used to store and manage important information such as user details, pet details, adoption requests, and volunteer applications.
- **Server-side Processing:** JSP and JDBC are used for processing forms, managing sessions, and communicating with the MySQL database.

The project is mainly focused on providing a simple, organized, and user-friendly online platform for pet adoption management. It can be further extended in the future with features such as secure password encryption, email notifications, online donation facilities, advanced pet search and recommendation, and a mobile application.

1.3 Objective

The main objective of the PetNest Pet Adoption Management System is to develop a simple, reliable, and user-friendly web-based platform that makes the process of pet adoption more organized and accessible. The system is designed to provide a common platform where users can find information about pets available for adoption, understand the adoption process, submit adoption requests, and participate in volunteer activities.

The project aims to reduce the difficulties involved in maintaining pet and adoption-related information manually. By using a centralized database, the system allows important records such as user details, pet information, adoption requests, and volunteer applications to be stored and managed systematically. This helps the organization maintain records efficiently and makes the information easier to access whenever required.

The major objectives of the PetNest project are:

To provide an online pet adoption platform where users can easily explore pets available for adoption and obtain the necessary information before making an adoption decision.

To provide detailed pet information, including the pet's name, breed, age, description, and image, so that users can understand the basic details of the pet.

To simplify the adoption process by allowing registered users to submit adoption requests online instead of depending completely on manual forms or physical communication.

To provide user registration and login functionality so that users can create accounts and access the features available to registered users.

To provide a volunteer application facility through which interested users can apply to become volunteers and provide information about their availability and reason for volunteering.

To develop an Admin Panel that allows administrators to manage important system records such as users, pets, adoption requests, and volunteer applications.

To manage adoption and volunteer applications efficiently by allowing administrators to review applications and update their status according to the organization's requirements.

To maintain centralized and organized data using a MySQL database so that user, pet, adoption, and volunteer information can be stored and retrieved easily.

To reduce manual record keeping and paperwork by providing digital forms and database-based management of adoption and volunteer information.

To provide useful information about PetNest, including its mission, events, gallery, success stories, adoption guidance, and contact details.

To provide a simple AI Assistant that can give quick answers to frequently asked questions related to pet adoption, volunteering, login, events, gallery, contact information, and other basic website-related queries.

To provide a user-friendly interface with simple navigation so that users can easily access different sections of the website.

To demonstrate the practical use of web technologies such as HTML, CSS, JavaScript, JSP, JDBC, MySQL, and Apache Tomcat in developing a complete web-based application.

To create a system that can be further enhanced in the future with additional features such as email notifications, secure password management, advanced pet searching and recommendation, online donation facilities, and mobile application support.

Overall, the objective of PetNest is to create an organized digital platform that supports the pet adoption process, improves record management, provides useful information to users, and helps administrators manage adoption-related activities efficiently.

2. System Analysis and Specification

2.1 Existing System

In the existing system, information about pets available for adoption is usually shared through different methods such as social media, phone calls, personal visits, or manual records. People who want to adopt a pet may have to contact the organization to get information about available pets and the adoption process.

The adoption and volunteer information may also be maintained manually in notebooks, forms, or separate files. Because of this, it can be difficult to keep all the information in one place and manage applications properly.

There may not be a single platform where users can view pet details, submit adoption requests, apply for volunteering, and get other information about the organization.

The existing system mainly depends on **manual communication and record keeping**, which can take more time for both users and administrators.

2.2 Limitations of Existing System

The existing system has some limitations:

1. **Manual Record Keeping:** Pet and application information may be maintained manually, which can be difficult to manage.
2. **Time Consuming:** Users may need to call or personally visit the organization to get information.
3. **Scattered Information:** Information about pets, adoption, volunteering, and events may be available through different sources.
4. **Difficult Application Tracking:** It can be difficult to track the status of adoption and volunteer applications manually.

5. **More Paperwork:** Manual forms and records can increase paperwork and require additional storage.
6. **Limited Accessibility:** Users may not be able to access pet information anytime from a single platform.
7. **Repeated Questions:** Administrators may need to answer the same basic questions about pets, adoption, contact details, and volunteering repeatedly.
8. **Difficulty in Managing Data:** As the number of users and applications increases, managing records manually becomes more difficult.
9. **No Centralized Database:** There may be no single database to store and manage all important information.
10. **Limited User Interaction:** Users may not have an easy online way to interact with the organization and submit their requirements.

2.3 Feasibility Study

A feasibility study is done to check whether the proposed **PetNest system** can be developed and used successfully. The project is studied from different points of view

2.3.1 Technical Feasibility

The project is technically feasible because it uses common and easily available technologies such as **HTML, CSS, JavaScript, JSP, JDBC, MySQL, and Apache Tomcat**.

These technologies can work together to create the required website, database, login system, adoption forms, volunteer forms, and Admin Panel. The project can also be developed and tested on a normal computer.

Therefore, the system is technically possible to develop and maintain.

2.3.2 Economic Feasibility

The project is economically feasible because the main technologies used for development are freely available or commonly used for academic projects.

The system does not require expensive hardware or software for basic development and testing. It can be developed using a local computer with MySQL and Apache Tomcat.

Therefore, the development cost of the system is low.

2.3.3 Operational Feasibility

The system is easy to operate because it provides simple navigation and user-friendly forms. Users can register, log in, view pets, submit adoption requests, and apply for volunteering.

Administrators can use the Admin Panel to manage users, pets, adoption requests, and volunteer applications.

Therefore, the system is operationally feasible and can be used by users and administrators with basic computer knowledge.

2.3.4 Schedule Feasibility

The project can be developed within the available academic project period by dividing the work into different modules such as website design, database creation, login system, adoption module, volunteer module, Admin Panel, AI Assistant, testing, and documentation.

Therefore, the project is also feasible from a time point of view.

2.4 Need of New System

The new **PetNest Pet Adoption Management System** is needed to overcome the problems of the existing manual system. The main purpose of the new system is to provide one common platform where users can easily access pet adoption-related information and administrators can manage the records.

The need for the new system is as follows:

1. To provide **one platform** for pet adoption-related information.
2. To allow users to **view available pets online** without needing to visit the organization first.
3. To provide an easy way to **submit adoption requests online**.
4. To allow interested people to **apply for volunteer opportunities** online.
5. To store user, pet, adoption, and volunteer information in a **centralized database**.
6. To make it easier for administrators to **manage users and applications**.
7. To allow administrators to **approve or reject applications** and maintain their status.
8. To reduce **manual paperwork and record keeping**.
9. To save time for both users and administrators.
10. To provide important organization information such as **About Us, Events, Gallery, Success Stories, and Contact details** in one place.
11. To provide a simple **AI Assistant** that can answer common questions quickly.
12. To make the adoption process more **organized, accessible, and convenient**.

Overall, the new system helps replace a mostly manual process with a simple web-based system that can store and manage information more efficiently.

2.5 Front End & Back End Tools

<i>Component</i>	Technology	Purpose
Front End	HTML	Page structure and forms
Styling	CSS	Layout, colors, responsive presentation
Client-side Logic	JavaScript	Validation, filtering, modals, AI Assistant
Server-side	JSP	Dynamic pages and form processing
Database Connectivity	JDBC	Connection between JSP and MySQL
Database	MySQL	Persistent storage of project records
Web Server	Apache Tomcat	Runs JSP application
Development	VS Code / similar IDE	Editing and debugging project files

Front End

The front end of PetNest is developed using **HTML, CSS, and JavaScript**.

- **HTML** is used to create the structure of the website.
- **CSS** is used for colors, layouts, buttons, forms, navigation bars, and overall design.
- **JavaScript** is used for interactive features such as pet filtering, form-related actions, modals, and the simple AI Assistant.

Back End

The back end of PetNest is developed using **JSP, JDBC, MySQL, and Apache Tomcat**.

- **JSP** is used to process user requests and create dynamic web pages.
- **JDBC** is used to connect the JSP pages with the MySQL database.
- **MySQL** stores important data such as users, pets, adoption requests, and volunteer applications.
- **Apache Tomcat** is used as the web server to run the JSP-based application.

This combination provides a simple and suitable environment for developing the PetNest web application.

3. System Requirement Specification

3.1 Proposed System and Advantages

The proposed **PetNest Pet Adoption Management System** is a web-based system designed to make the pet adoption process simple and organized. It provides a common platform where users can view available pets, learn about their details, submit adoption requests, and apply for volunteer opportunities.

The system also provides an **Admin Panel** through which the administrator can manage users, pets, adoption requests, and volunteer applications. All important information is stored in a MySQL database, which makes it easier to maintain and retrieve records.

The system also includes a simple **AI Assistant** that provides quick answers to common questions related to pet adoption, volunteering, contact information, events, and other website features.

The proposed system provides the following main functions:

- User registration and login.
- Viewing and searching available pets.
- Viewing detailed information about pets.
- Submitting online adoption requests.
- Applying for volunteer opportunities.
- Managing users through the Admin Panel.
- Managing pet information.
- Managing adoption requests.
- Managing volunteer applications.

- Viewing events, gallery, and success stories.
- Providing organization and contact information.
- Answering common questions through the AI Assistant.
- Storing important information in a MySQL database.

Overall, the proposed system provides a **single platform for users and administrators** to manage and access pet adoption-related information. It helps make the adoption process more convenient for users and makes record management easier for the administrator. The system can also be extended in the future by adding more features according to the requirements of the organization.

- ☐ **Easy Pet Adoption:** Users can easily view pet details and submit adoption requests through the website.
- ☐ **User-Friendly Interface:** The system provides a simple and easy-to-use interface for users and administrators.
- ☐ **Online Volunteer Application:** Users can apply for volunteer opportunities online without any manual paperwork.
- ☐ **Centralized Data Management:** User, pet, adoption and volunteer information is stored and managed systematically using a MySQL database.
- ☐ **Admin Management:** The administrator can manage users, pets, adoption requests, volunteer applications and other website information from the Admin Panel.
- ☐ **AI Assistant:** The integrated AI Assistant helps users by providing quick answers to their questions related to the website and its features.
- ☐ **Time Saving:** The online system reduces manual work and makes the adoption and volunteer application process faster and more convenient.

Advantages of Proposed System

The main advantages of the proposed PetNest system are:

1. **Easy to Use:** The system provides simple navigation and easy-to-understand forms.
2. **Saves Time:** Users can view pet information and submit applications online without depending completely on physical visits.
3. **Centralized Data:** User, pet, adoption, and volunteer information can be stored in one database.
4. **Less Paperwork:** Online forms reduce the need for maintaining manual records.
5. **Easy Application Management:** Administrators can easily view and manage adoption and volunteer applications.
6. **Quick Information:** Users can easily access information about pets, events, volunteering, and contact details.
7. **AI Assistant:** The simple AI Assistant provides quick answers to frequently asked questions.
8. **Better Record Management:** Database storage makes it easier to find and manage records.
9. **User Convenience:** Users can access the website and submit requests from their own devices.
10. **Future Expandability:** The system can be extended with features such as notifications, online donations, advanced pet searching, and mobile applications.

3.2 Development Strategy Model

For developing the **PetNest** project, an **Incremental Development Model** can be used. In this approach, the complete system is divided into smaller modules, and each module is developed and tested step by step.

Instead of developing the complete system at once, different features are completed one by one. After completing and testing one module, the next module is developed and integrated with the existing system.

Development Phases

1. Requirement Analysis

First, the requirements of the project are identified. The main requirements include user registration, login, pet browsing, adoption requests, volunteer applications, Admin Panel, database management, and AI Assistant.

2. System Design

The structure of the website, database, navigation, forms, and Admin Panel is planned. The required database tables and relationships are also designed.

3. Front-End Development

The website interface is developed using **HTML, CSS, and JavaScript**. Different sections such as Home, Pets, Volunteer, Events, Gallery, About, and Contact are designed.

4. Database Development

MySQL is used to create and manage the database. Tables are created for storing users, pets, adoption requests, and volunteer applications.

5. Back-End Development

JSP and JDBC are used to process forms and connect the website with the MySQL database. Login, registration, adoption, volunteer, and administrative functions are implemented.

6. Admin Panel Development

The Admin Panel is developed to allow administrators to manage users, pets, adoption requests, and volunteer applications.

7. AI Assistant Integration

A simple JavaScript-based AI Assistant is added to provide predefined answers to common PetNest-related questions.

8. Testing

Each module is tested separately to find and correct errors. After that, the complete system is tested to check whether all modules work properly together.

9. Final Integration and Documentation

All completed modules are combined into the final system. The project is then documented through diagrams, screenshots, coding details, testing results, and the final project report.

Simple Development Flow

Requirement Analysis → System Design → Front-End Development → Database Development → Back-End Development → Admin Panel → AI Assistant → Testing → Final Integration

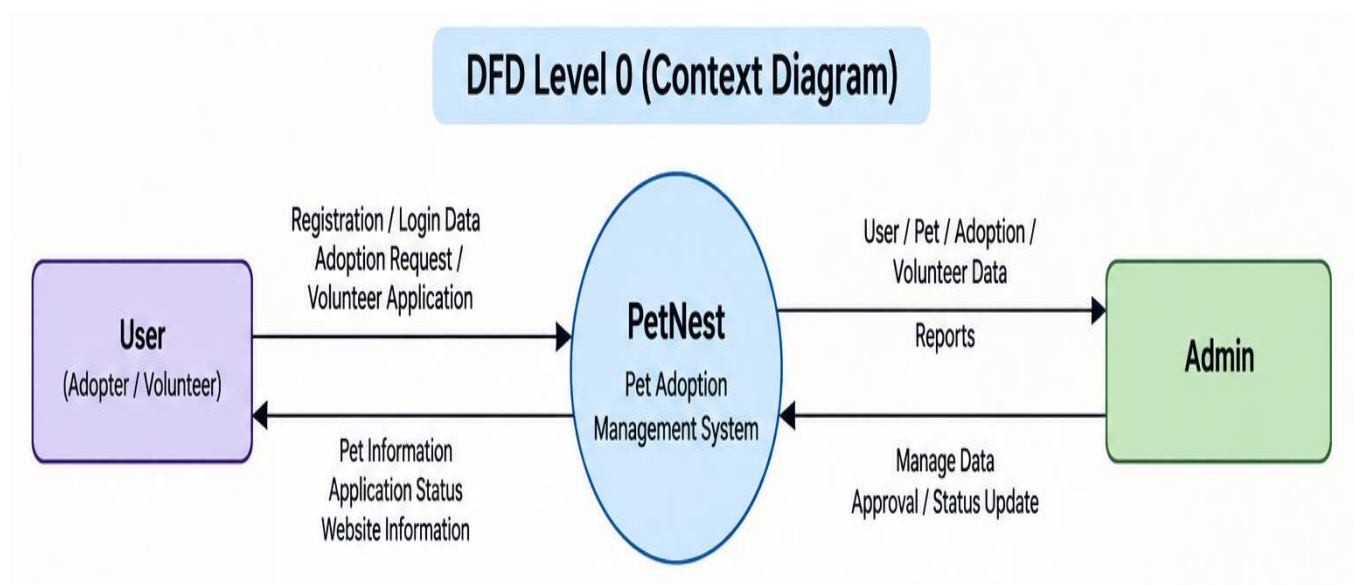
This development strategy makes the project easier to develop, test, and maintain because each module can be handled separately before integrating it into the complete system

4. System Design

4.1 Data Flow Diagrams

DFDs visualize how data moves through the system, focusing on data processing and interactions between users and the system.

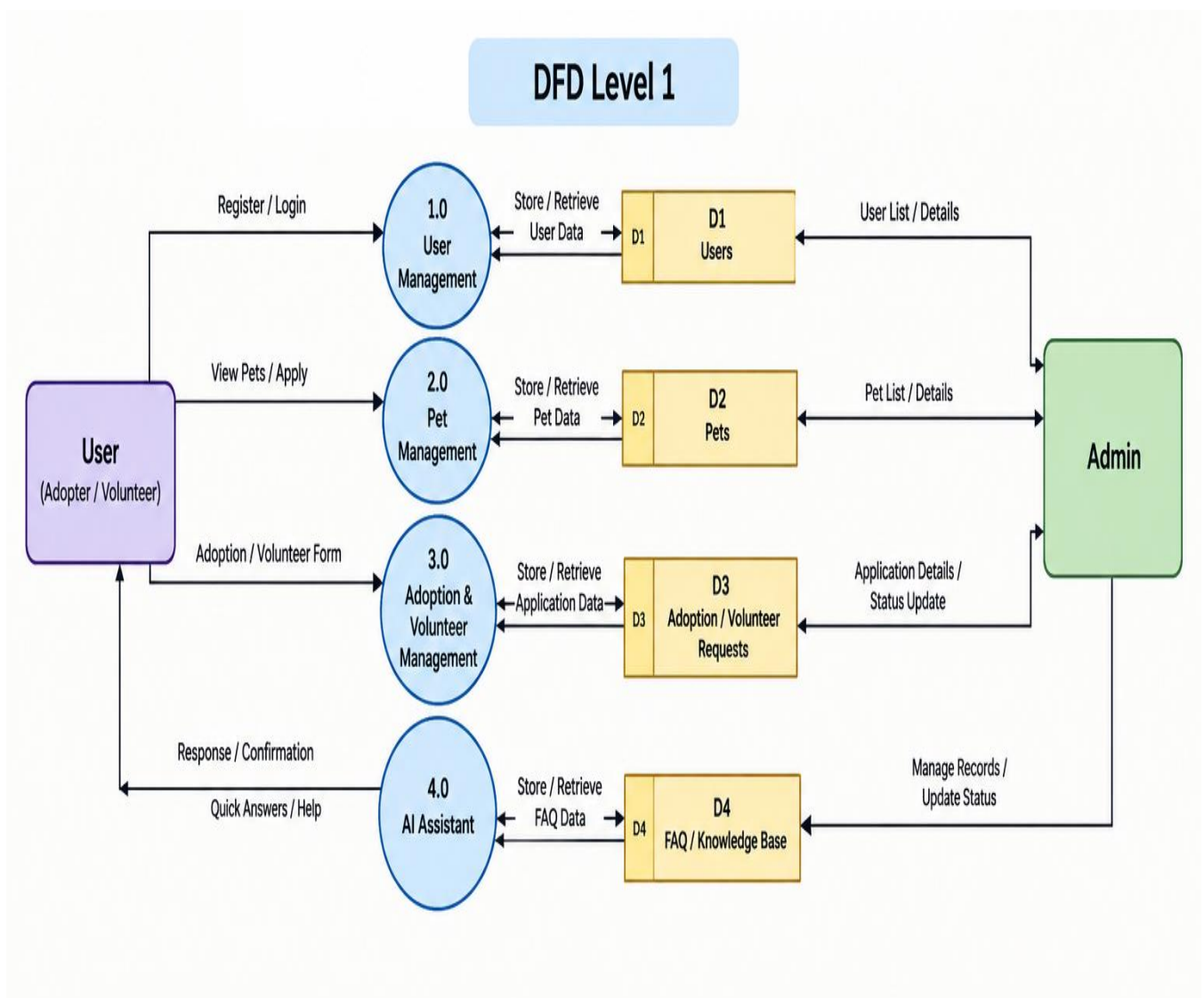
4.1.1 Level-0 DFD



Level-0 Data Flow Diagram

4.1.2 Level-1 DFD

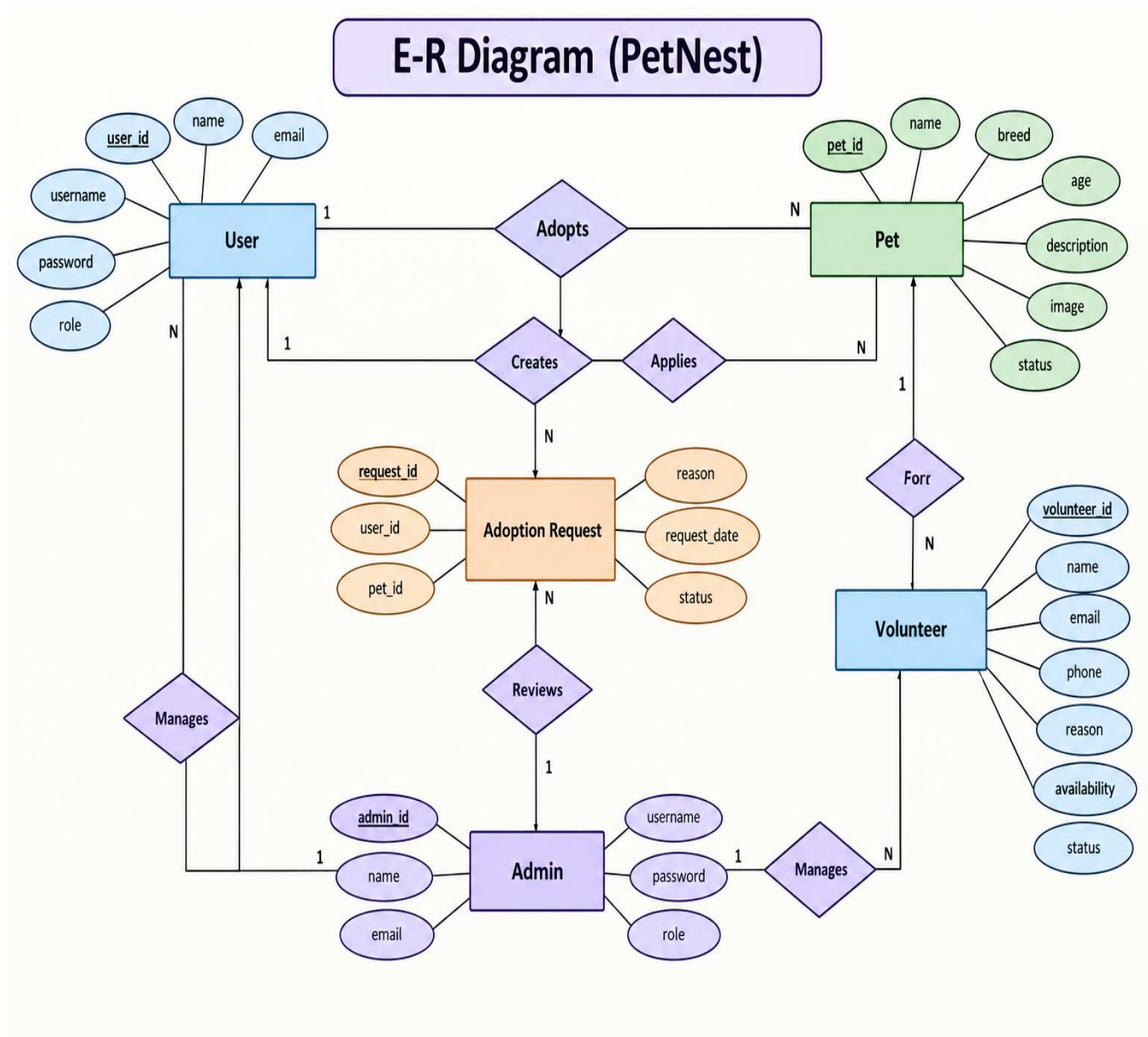
The Level-1 DFD decomposes the application into authentication, pet browsing, application processing, volunteer processing, and administrative management. These modules communicate with the MySQL database.



Level-1 Data Flow Diagram

4.2 E-R Diagram

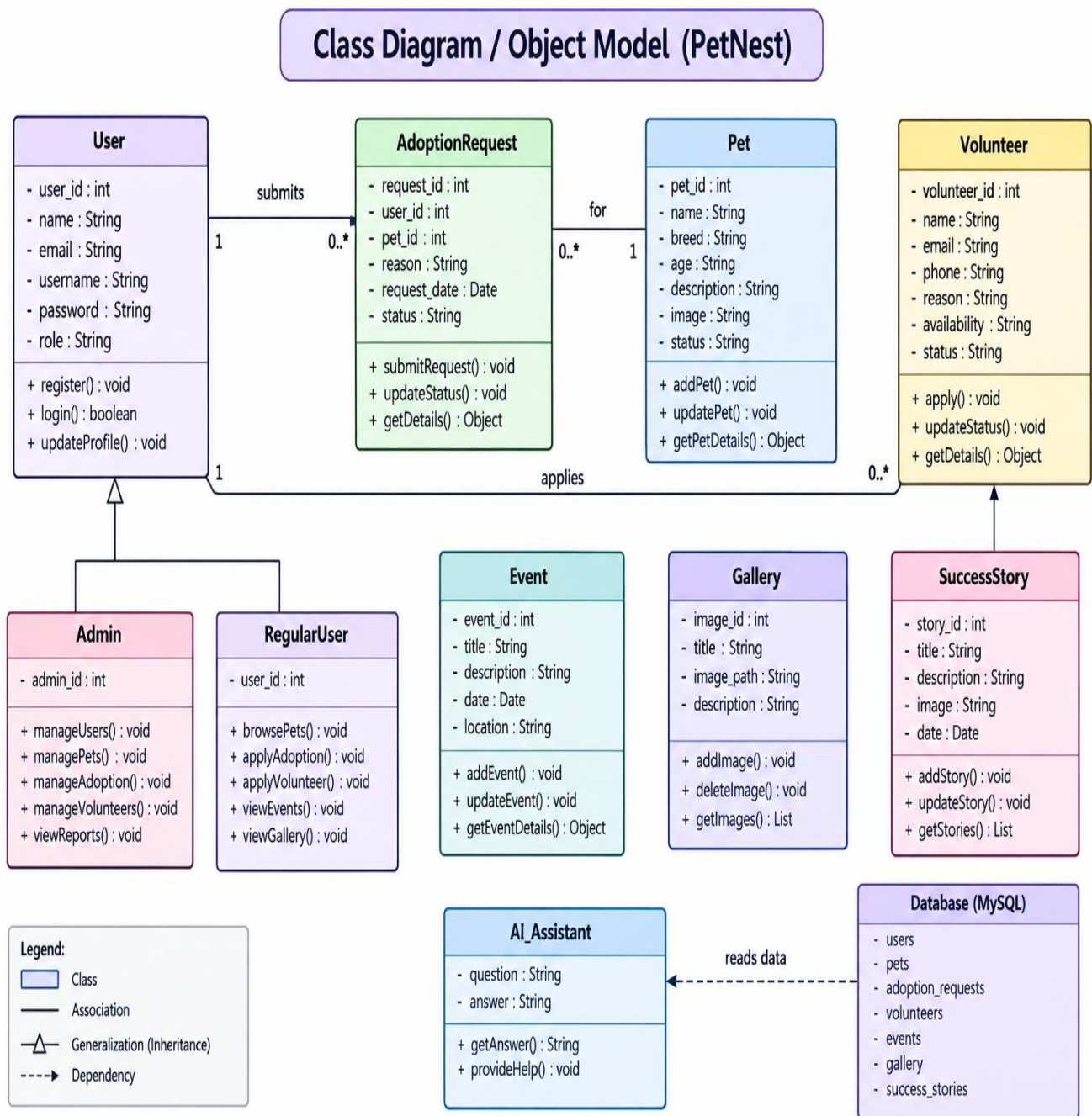
The main database entities are Users, Pets, Adoption Requests, and Volunteers. User records support authentication, pet records represent animals shown by the website, adoption requests store applications, and volunteer records store volunteer applications and their statuses.



Entity-Relationship Overview

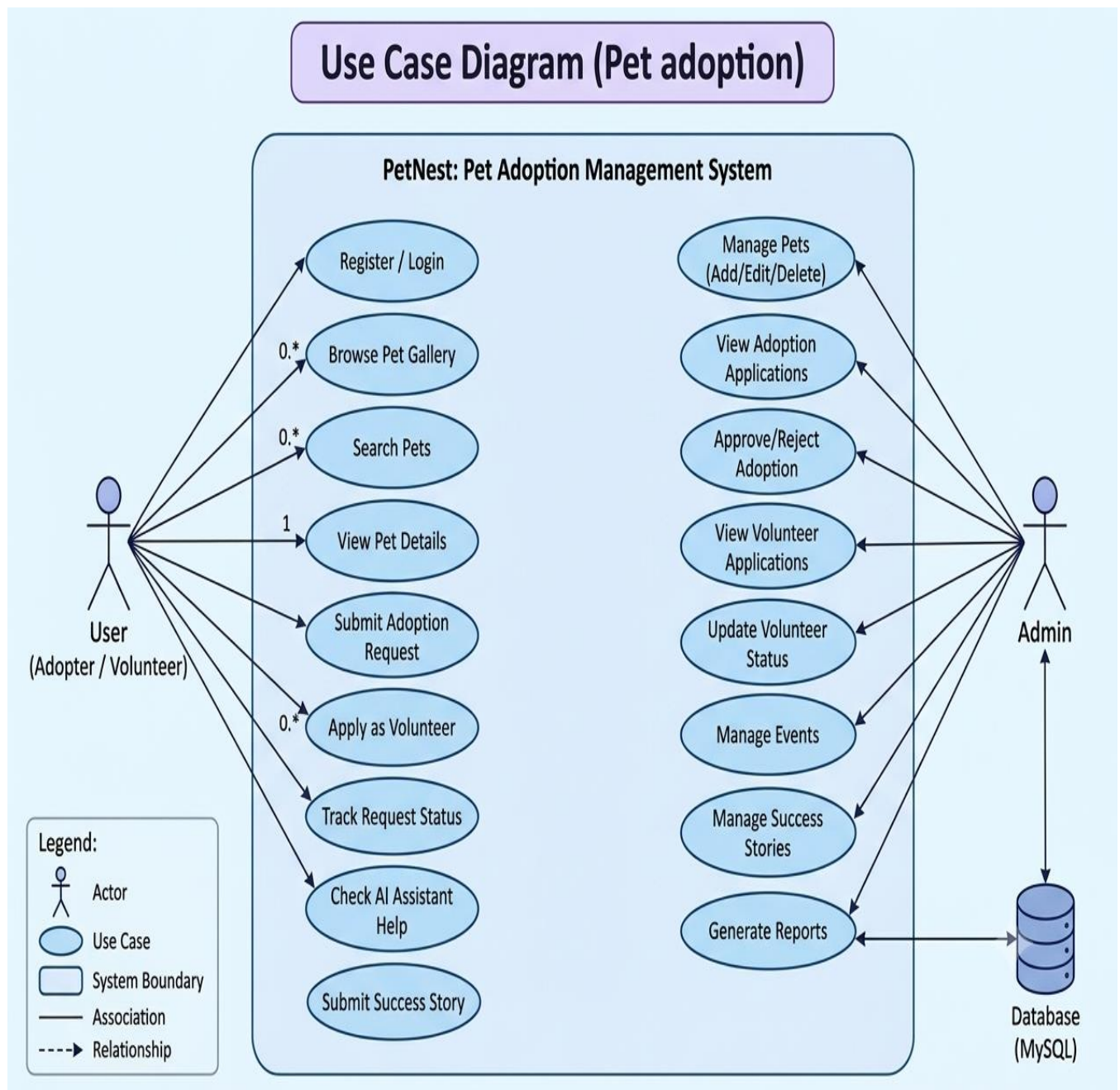
4.3 Class Diagram / Object Model

The project is implemented primarily with JSP pages rather than a large Java class hierarchy. Therefore, the logical object model can be represented through the main application entities and their operations. User and Admin represent actors, while Pet, AdoptionRequest, Volunteer, and DatabaseConnection represent core application concepts.



4.4 Use Case Diagram

The main user use cases are registration, login, browsing pets, submitting adoption requests, applying as a volunteer, and viewing public information. Administrator use cases include user management, adoption-request management, volunteer management, and pet management.



Use Case Diagram

4.5 Sequence Diagram

The sequence below represents a simplified adoption workflow. The user sends information through the browser, JSP processes the request, JDBC communicates with MySQL, and the administrator later reviews the stored request.

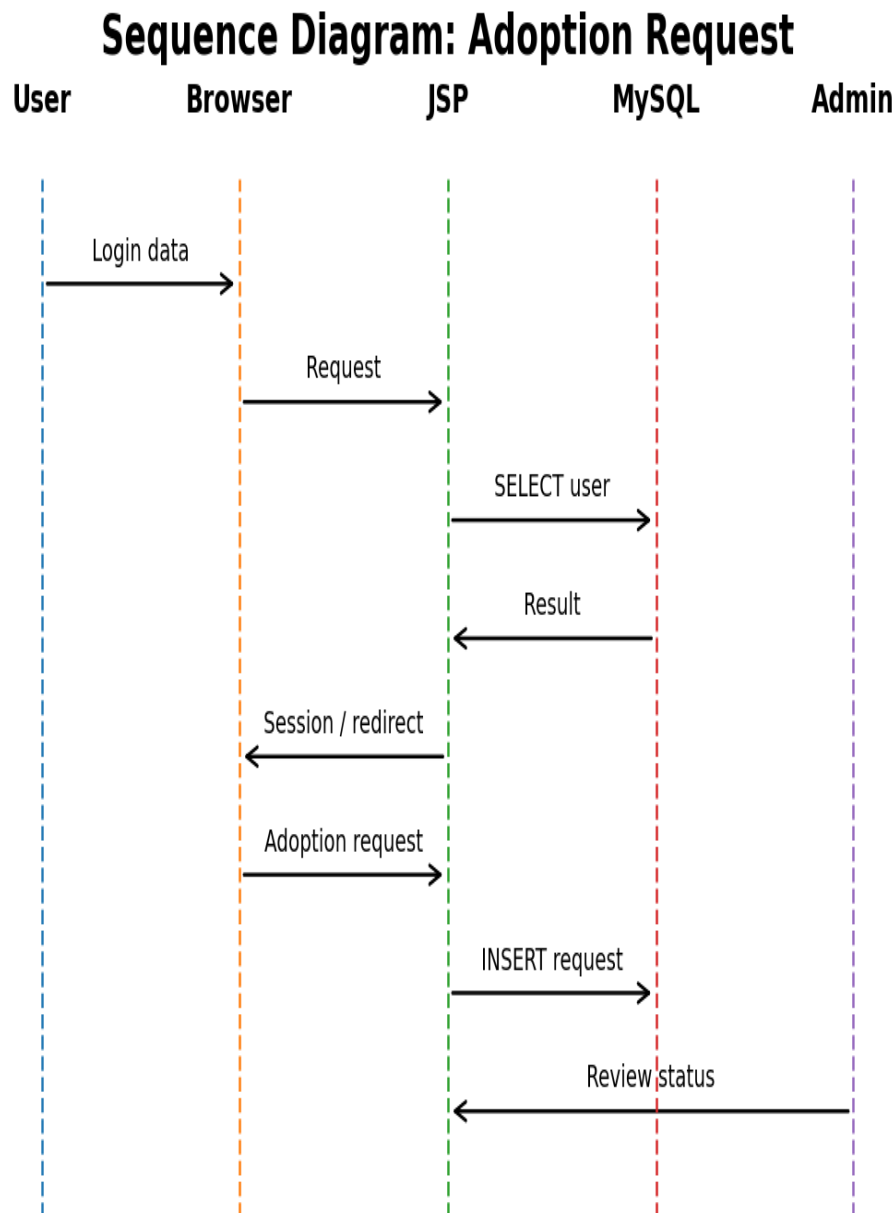
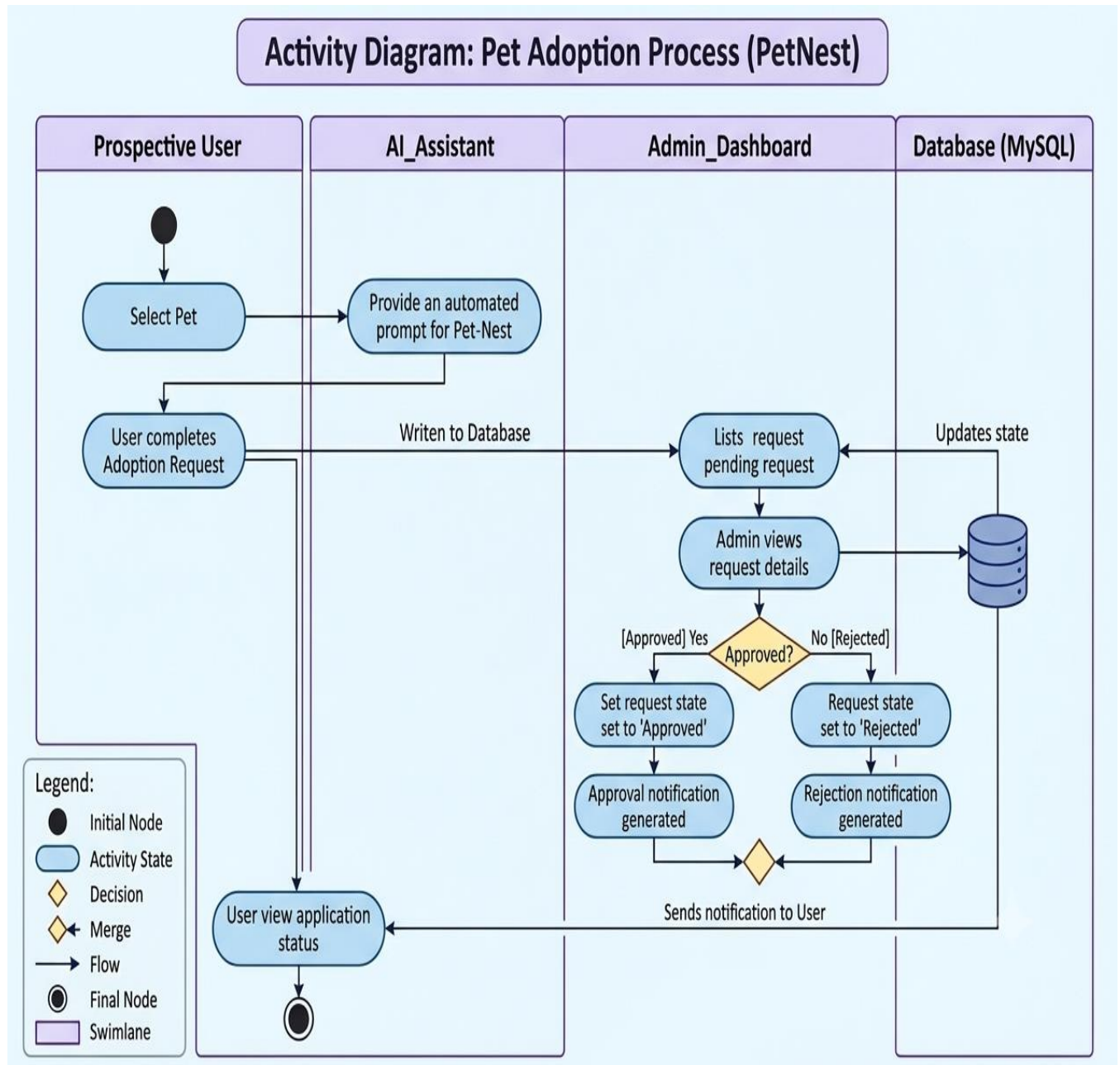


Figure 4.6: Adoption Request Sequence

4.6 Activity Diagram

The activity flow starts when a user opens PetNest, logs in or registers, browses a pet, and submits an adoption request. The request is stored in the database and becomes available to the administrator for review.



Pet Adoption Activity Diagram

4.7 Data Dictionary (Database Design)

4.8.1 Users Table

#	Name	Type	Null	Extra
1	Id	Int(11)	No	Primary key
2	Name	Varchar(100)	No	
3	Email	Varchar(100)	No	
4	Password	Varchar(255)	No	
5	Create at	Timestamp	No	

4.8.2 Pets Table

#	Name	Type	Null	Extra
1	Id	Int(11)		Primary key
2	Name	Varchar(255)		
3	Breed	Varchar(100)		
4	Age	Enum('puppy', 'adult', 'senior')		
5	Description	Text		
6	Image	Varchar(255)		

4.8.3 Adoption Requests Table

#	Name	Type	Null	Extra
1	Id	Int(11)	No	Primary key
2	Pet_name	Vacrchar(255)	No	
3	Pet_breed	Varchar(255)	No	
4	Pet_age	Varchar(50)	No	
5	Reason	Text	Yes	
6	Request_date	Timestamp	No	
7	Status	Varchar(20)	No	

4.8.4 Volunteers Table

#	Name	Type	Null	Extra
1	Id	Int(11)		Primary key
2	Name	Varchar(100)		
3	Email	Varchar(100)		
4	Phone	Varchar(15)		
5	Reason	Text		
6	Availability	Varchar(50)		
7	Status	Enum('pending', 'approved', 'reject')		

5. Screen Design

5.1 Input Design

Input design focuses on collecting only the information required for each operation. Forms use clear labels and simple fields so that users can complete them with minimal effort.

Screen	Input Fields	Purpose
Registration	Name, Email, Password	Create a user account
Login	Email, Password	Authenticate user
Adoption Form	Pet name, breed, age, reason	Submit adoption request
Volunteer Form	Name, Email, Phone, Reason, Availability	Apply as volunteer
Contact Form	Name, Email, Message	Send organization enquiry
Admin Login	Admin credentials	Protect administration area

5.2 Output / Report Design

Output screens display records in a readable format. The admin dashboard presents counts for users, pending adoption requests, and approved adoption requests. Management pages display records in tables with action buttons for status changes or deletion.

5.2.1 Main Website Screens

- Home: introduces PetNest and provides navigation to major sections.
- Browse Pets: displays available pet information and filters.
- Pet Details: shows selected pet information and adoption option.
- Volunteer: provides volunteer information and application form.
- Events and Gallery: provide community-event information and visual content.
- About Us: explains the purpose and mission of PetNest.
- Contact Us: presents organization contact information and enquiry form.
- Login / Register: supports user account creation and authentication.
- AI Assistant: floating question-answer interface for common PetNest queries.

5.2.2 Administrative Screens

- Admin Dashboard: summary counts and navigation.
- Manage Pets: lists pet records and provides management actions.
- Manage Users: displays registered users.
- Manage Adoption Requests: shows applications with Approve/Reject actions.
- Volunteer Applications: displays volunteer submissions and status actions.
- Manage Contact Messages: intended for administrative enquiry management.

Screenshot of the System

➤ Home page:

Home Browse Pets Volunteer Events Gallery How to Adopt Success Stories About Us Contact Us Welcome: p@p.p Logout

WELCOME TO PETNEST

Your one-stop destination to find and adopt your next best friend.

Browse Pets

AVAILABLE PETS FOR ADOPTION

All Pets All Ages

Buddy
Breed: Indian pariah
Age: Adult
[View Details](#)

Whiskers
Breed: Maine Coon
Age: Adult
[View Details](#)

Rocky
Breed: Beagle
Age: Puppy
[View Details](#)

Lucy
Breed: Parrot
Age: Adult
[View Details](#)

Be A Volunteer

Join us in making a difference in the lives of pets. Fill out the form below to volunteer!
Check Your Status if you already fill Up the Form [Check](#)

Your Name:
Enter your name

Your Email:
Enter your email

Phone Number:
Enter your phone number

Why do you want to be a Volunteer?
Please tell us why you want to be a Volunteer

Availability
Select availability

Submit Application

Upcoming Events

Join our events to support animal welfare and make a difference.

Adoption Drive
Date: February 10, 2025
Location: Central Park
Description: A day to find loving homes for our pets. Come and meet your furry friend!

Fundraiser Gala
Date: March 15, 2025
Location: Grand Hotel
Description: An evening of dining and entertainment to raise funds for animal welfare.

Past Event Gallery

Browse through the highlights of our past events.

© 2025 PetNest. All rights reserved. Made with ❤️ by Payal

Description :

Purpose of the PetNest Website

➤ **Promote Pet Adoption:**

Help users find and adopt pets by displaying available animals with details like breed, age, and images.

Provide filters to search for specific types of pets.

➤ **Encourage Volunteering:**

Allow people to apply as volunteers for adoption events and other activities through an online form.

➤ **Share Events and Success Stories:**

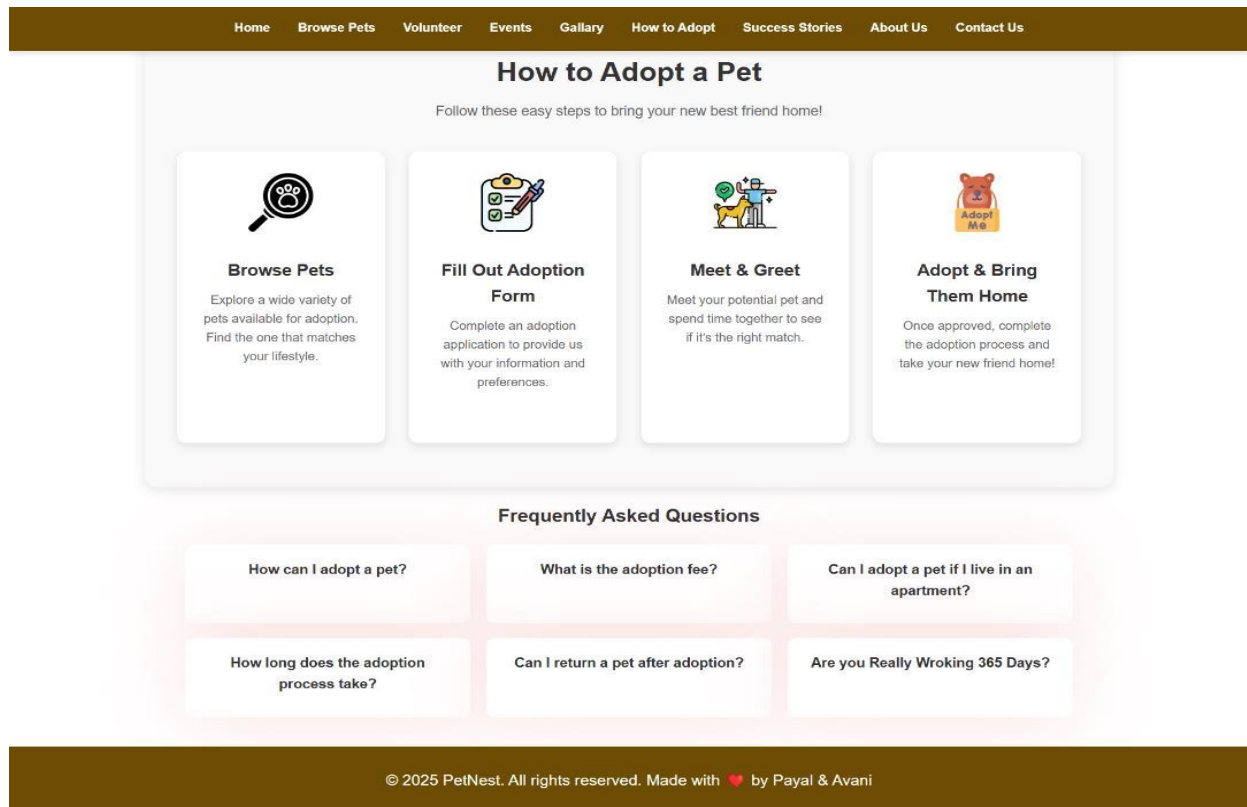
Showcase upcoming events to promote pet welfare (e.g., adoption drives, fundraisers, health camps).

Display past event photos and success stories of adopted pets.

➤ **Simplify Pet Adoption:**

Provide easy navigation to view pet details and contact the organization for adoption inquiries.

➤ How to Adopt Page:



Description:

The "How to Adopt a Pet" section provides a simple, step-by-step guide to adopting a pet. It outlines the following four key steps:

1. *Browse Pets*

- Explore a variety of pets available for adoption.
- Find a pet that matches your lifestyle and preferences.

2. *Fill Out Adoption Form*

- Complete an adoption application.

- Provide your information and preferences to help find the best match.

3. *Meet & Greet*

- Meet your potential pet in person.
- Spend time together to ensure compatibility.

4. *Adopt & Bring Them Home*

- Once approved, complete the adoption process.
- Take your new pet home and start your journey together!

This section makes it easy for users to understand the adoption process and take the necessary steps to welcome a pet into their home.

➤ Success Stories page:

[Home](#)
[Browse Pets](#)
[Volunteer](#)
[Events](#)
[Gallery](#)
[How to Adopt](#)
[Success Stories](#)
[About Us](#)
[Contact Us](#)

Heartwarming Adoption Stories



Adopted: Bella (Dog)

"When I first saw Bella at the shelter, I knew she was meant to be part of my life. She was a shy little pup, hesitant to approach strangers, but the moment she wagged her tail at me, I felt an instant connection. Bringing Bella home was the best decision I've ever made. She has filled my days with joy, laughter, and unconditional love. She loves her evening walks, enjoys playing fetch in the park, and has the cutest way of resting her head on my lap when she's sleepy. Bella is not just a pet; she's family."

- Rohit Ahuja



Adopted: Luna (Cat)

"Luna was found as a stray, scared and hungry. The moment I saw her big, curious eyes staring up at me, I knew I had to take her home. Over time, she blossomed into the most affectionate and playful cat I've ever known. She loves chasing laser pointers, curling up on my lap while I read, and greeting me at the door after a long day. Adopting Luna has brought so much warmth and happiness into my life. I can't imagine my home without her!"

- Emily Smith



Adopted: Rio (Parrot)

"When I adopted Rio, I had no idea how much he would change my life. Every morning, he greets me with a cheerful 'Good morning!' and throughout the day, he keeps me entertained with his funny mimicry. He loves sitting on my shoulder while I work and enjoys singing along to music. His vibrant personality and loving nature make every day brighter. He may have come from a rescue shelter, but in truth, he rescued me with his endless joy and companionship."

- Rhea Sharma



Adopted: Charlie (Cat)

"Charlie came into my life when I needed a friend the most. From the moment he stepped into my home, he claimed every corner as his own. He loves sitting by the window, watching the world go by, and curling up beside me at night. His gentle purring is the most comforting sound. He has brought so much love and happiness into my life, and I am grateful every day for the companionship he gives me."

- Sophia Wilson



Adopted: Rocky (Dog)

"Rocky is not just my pet; he's my adventure buddy! Whether it's hiking through the mountains, running on the beach, or simply playing in the backyard, he's always ready for an adventure. He has a heart full of love and an endless supply of energy. I adopted him when he was just a pup, and watching him grow into the loving and loyal dog he is today has been the most rewarding experience. Rocky has made my life so much more exciting and fulfilling!"

- Gracy Oberoi



Adopted: Milo (Parrot)

"Milo is the most entertaining companion I could have ever asked for. When I first brought him home, he was shy and quiet, but now he fills my house with laughter. He loves mimicking my voice, whistling songs, and even calling my name when he wants attention. He has taught me patience, responsibility, and the beauty of companionship. He's more than a pet—he's my feathered best friend!"

- Jessica Kohli



Adopted: Whiskers (Cat)

"Whiskers is a little ball of energy! He loves playing hide-and-seek, stealing my socks, and following me everywhere. I never expected a cat to be so affectionate, but Whiskers has proven me wrong. Whether I'm watching TV, working on my laptop, or just relaxing, he's always by my side. His playful nature and loving heart have turned every day into a joyful experience. He is truly my best friend!"

- Rose Pectus

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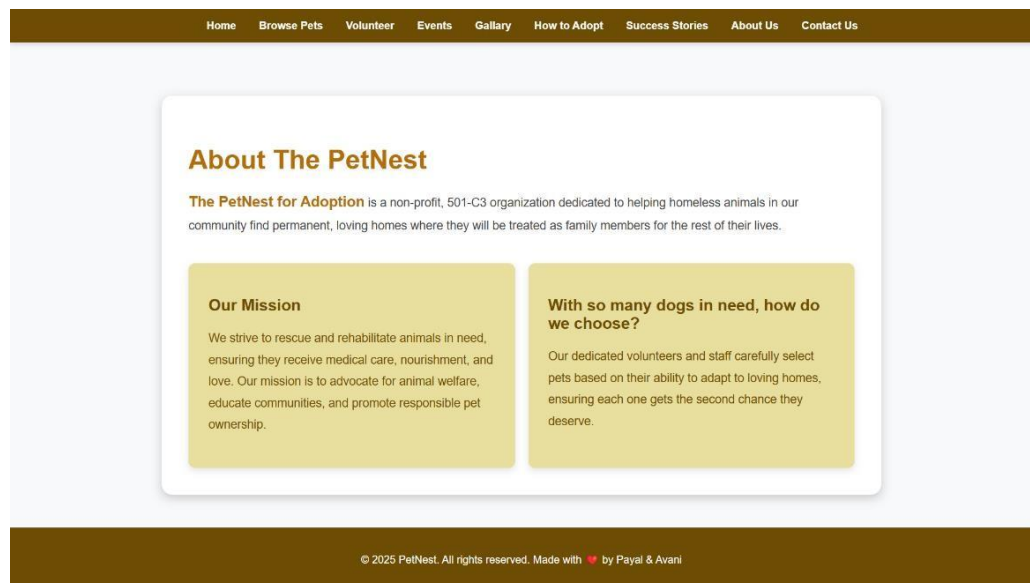
Description:

The purpose of this Heartwarming Adoption Stories page is to:

1. **Inspire Potential Adopters** – By sharing real-life success stories, the page encourages others to adopt pets by showcasing the joy and companionship they bring.
2. **Highlight the Positive Impact of Adoption** – The testimonials demonstrate how rescued pets transform lives, emphasizing the emotional bond between adopters and their pets.
3. **Build Trust in the Adoption Process** – Real stories from adopters reassure potential pet owners about the process and the benefits of adopting from a shelter.
4. **Celebrate Successful Adoptions** – It serves as a tribute to both pets and their new families, celebrating their journey to a loving home.
5. **Encourage More Adoptions** – Seeing happy pets in their forever homes may motivate visitors to consider adopting a pet themselves.

Overall, this page creates an emotional connection with visitors, reinforcing the idea that adopting a pet is a rewarding and life-changing experience.

➤ About Us Page:



Description:

Welcome to PetNest where every pet deserves a loving home!

We are a dedicated team of animal lovers committed to rescuing, rehabilitating, and rehoming abandoned, stray, and surrendered pets. Our mission is to provide these animals with a second chance at happiness by connecting them with caring families.

At PetNest we believe that adoption is the key to reducing pet homelessness. Through our compassionate care, medical support, and behavioral training, we ensure that every pet is ready to become a cherished member of your family.

➤ Contact us Page:

Description:

Our “Contact Us” page offers a simple and user friendly form for visitor to get in touch.

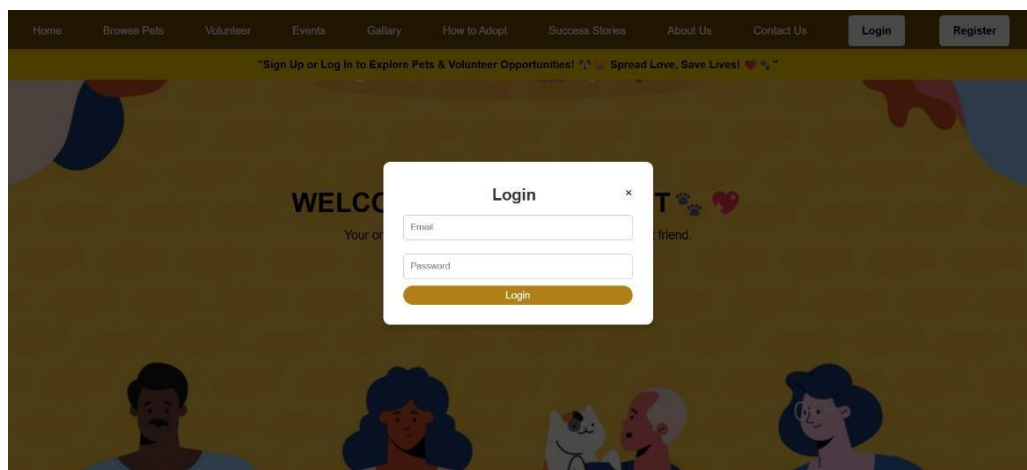
The form include three fields: name , email and message ,allowing user to easily provide their details and inquiries.

And also we can provide you our NGO address if any visitor want to come then they can easily reach to us.

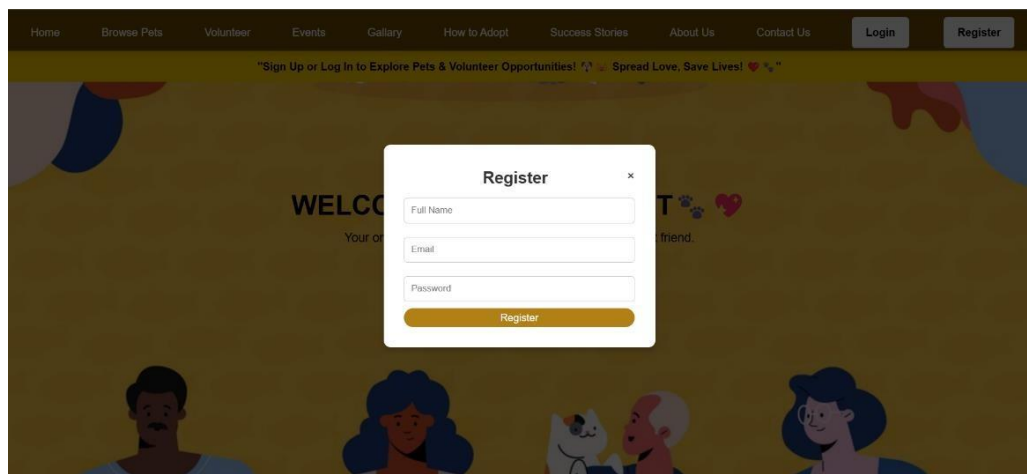
The Contact Us page allows users to communicate with the PetNest team by submitting their name, email address, and message. It provides a simple way for users to send enquiries, feedback, or other queries related to the pet adoption website.

➤ Login and Register page:

Login :



Register:



Description:

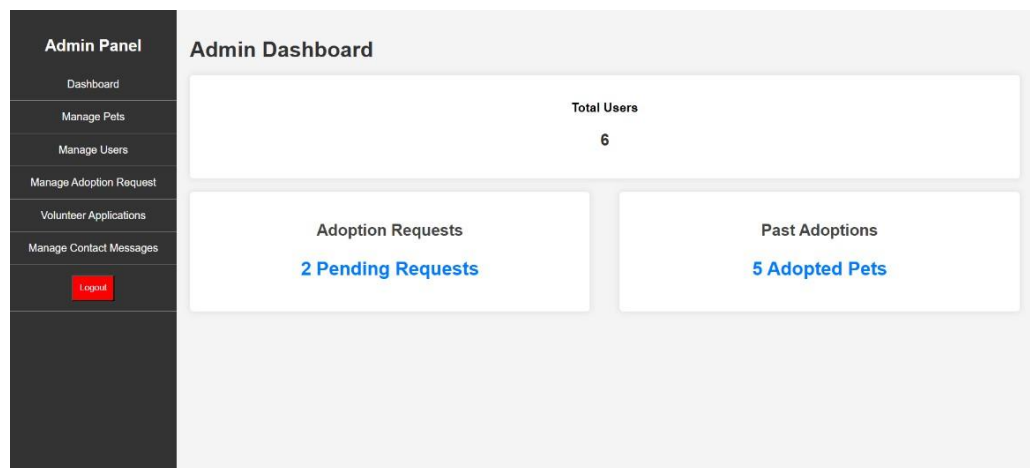
Login Screen:

This screen allows registered users and administrators to securely log in to the PetNest system using their login credentials.

Registration Screen:

This screen allows new users to create an account by entering the required personal and login details.

➤ Admin Panel Home Page:



Description:

The Admin Panel Dashboard is designed to allow website administrators to manage the entire system efficiently. It provides access to key functionalities such as managing pets, users, events, volunteers, and viewing the website's performance.

The PetNest Admin Panel is a comprehensive dashboard that provides full control over the pet adoption website. It enables efficient management of:

Pets, Users, Volunteers, Events, Gallery

This system ensures that the website is always updated and running smoothly, enhancing the user experience.

➤ Manage User Page:

[Go to Dashboard](#)

Manage Users			
ID	Name	Email	Action
1	payal	p@p.p	Delete
2	bhagu	b@b.b	Delete
3	avani	a@a.a	Delete
6	manu	m@m.m	Delete
7	yashvi	yashvi@gamil.com	Delete
8	jitakshi	j@j.j	Delete

Description:

The User Management module in the admin panel allows administrators to:

- View, manage, and control user accounts.
- Ensure only genuine and active users remain on the platform.
- Monitor volunteer participation.

➤ Manage Adoption Request:

[Go to Dashboard](#)

Manage Adoption Requests						
ID	Pet Name	Pet Breed	Pet Age	Reason	Status	Action
1	Buddy	Labrador	Adult	aaaa	Approved	Reject
2	Buddy	Labrador	Adult	love pets	Rejected	Approve
3	oreo	maine coon	Adult	love to take care of pets	Approved	Reject
4	cooper	Labrador	senior	love pets	Approved	Reject
5	tiger	rottweiler	puppy	love to take care of pets	Approved	Reject
6	buddy	dog	adult	love for pets	Approved	Reject
7	buddy	dog	adult	love for pets	pending	Approve Reject
8	buddy	dog	adult	love for pets	pending	Approve Reject

Description:

Purpose of Pet Adoption Request Management

The Manage Pet Adoption Requests module allows administrators to:

- View, approve, reject, or track adoption requests submitted by users.
- Ensure pets are being adopted by genuine users.
- Maintain accurate adoption records in the database.
- Keep the website's pet availability status up to date.

➤ Manage Contact Us Messages:

[Go to Dashboard](#)

Contact Us Messages				
ID	Name	Email	Message	Submitted At
4	soham	r@gmail.com	kjgolol	2025-03-28 12:12:29.0
3	soham	r@gmail.com	eglmjh	2025-03-27 12:05:56.0
2	kashyap	a@gmail.com	hgfh	2025-03-25 11:34:44.0
1	payal	p@p.p	hello	2025-03-24 11:53:02.0

Description:

The "Contact Us" messages module in the admin panel allows administrators to:

- View, respond to, and manage inquiries submitted through the website's Contact Us form.
- Keep track of user feedback, complaints, and questions.
- Ensure timely responses and maintain customer satisfaction.

Improve communication efficiency by organizing messages.

➤ Manage Volunteer Application Form:

[Go to Dashboard](#)

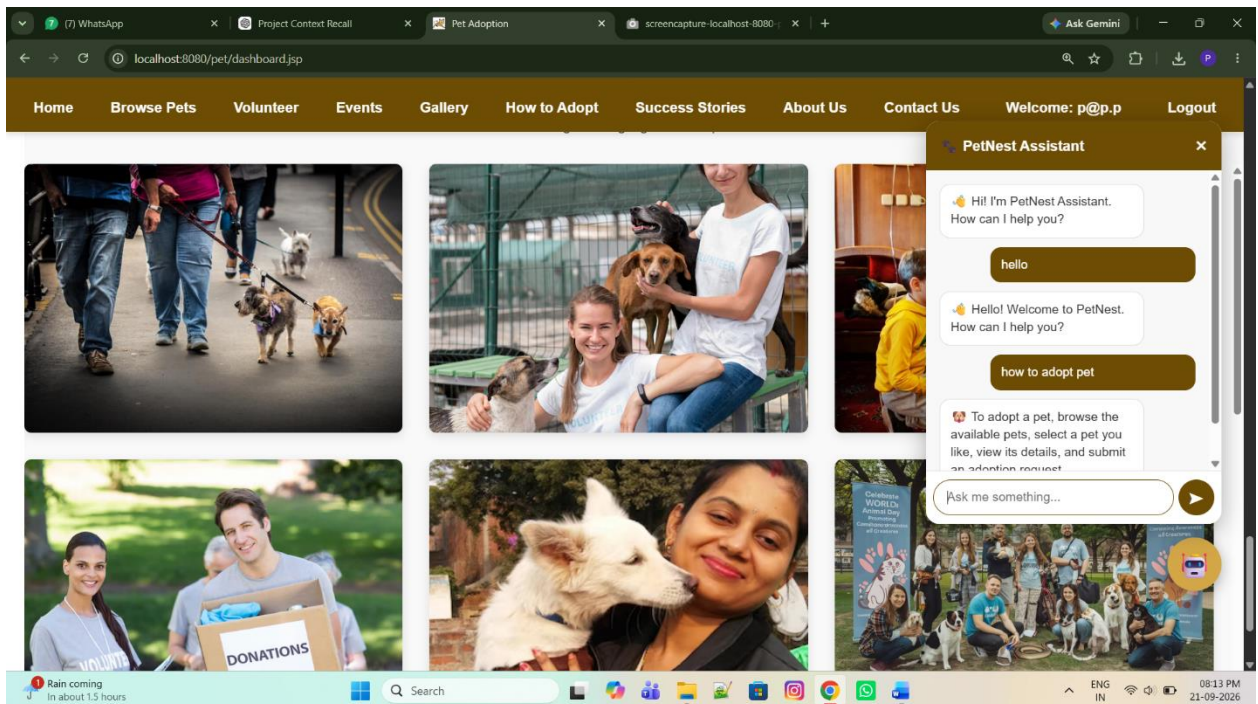
Manage Volunteers							
ID	Name	Email	Phone	Reason	Availability	Status	Action
5	payu	p@gmail.com	12345	aaa	flexible	Approved	Reject
6	avani	a@a.a	54545	i love pets	flexible	Approved	Reject
9	JIVANI MANASVI	manasvi@yahoo.com	1234567890	khfcmj	weekends	Approved	Reject
10	mansvi	manasvi@yahoo.com	1234567890	kjlf,k.o[	flexible	Pending	Approve Reject
11	mansvi	mans@gmail.com	1234567890	fbh	weekends	Rejected	Approve

Description:

The Volunteer Application feature in the Pet Adoption Website allows individuals to sign up as volunteers to assist with various tasks, such as:

- Caring for pets.
- Organizing adoption events.
- Helping with administrative work.
- Managing the website's content and pet listings.

➤ Chatbot:



Description:

This screen provides an AI-powered assistant where users can enter their questions and receive relevant answers. It helps users get quick information and guidance about the PetNest website, pet adoption, volunteering, and other available features.

6. Coding (Few Code Snippets)

6.1 Database Connection

JSP pages connect to MySQL using JDBC. A typical connection used in the project is:

```
Connection con = DriverManager.getConnection(
    "jdbc:mysql://localhost:3306/db", "root", ""
);
```

6.2 User Login

The login module uses a PreparedStatement to verify the submitted email and password and creates a session attribute after successful authentication.

```
PreparedStatement ps =
    con.prepareStatement(
        "SELECT * FROM users WHERE email=? AND password=?"
    );
ps.setString(1, email);
ps.setString(2, password);
ResultSet rs = ps.executeQuery();

if (rs.next()) {
    session.setAttribute("user", email);
}
```

6.3 Registration

Registration first checks whether the email already exists. If the email is not present, the user record is inserted into the users table.

```
PreparedStatement checkUser =
    con.prepareStatement("SELECT * FROM users WHERE email=?");

checkUser.setString(1, email);
ResultSet rs = checkUser.executeQuery();

if (!rs.next()) {
    PreparedStatement ps =
        con.prepareStatement(
            "INSERT INTO users (name,email,password) VALUES (?,?,?)"
        );
}
```

6.4 Adoption Request Processing

```
PreparedStatement stmt = con.prepareStatement(
    "INSERT INTO adoption_requests " +
    "(pet_name, pet_breed, pet_age, reason, request_date) " +
    "VALUES (?, ?, ?, ?, ?)"
);
```

6.5 Volunteer Application

```
String query =
    "INSERT INTO volunteers " +
    "(name, email, phone, reason, availability) " +
    "VALUES (?, ?, ?, ?, ?)";
```

```
PreparedStatement ps = con.prepareStatement(query);
```

6.6 PetNest AI Assistant

The AI Assistant is intentionally simple and does not call an external AI API. It uses JavaScript keyword matching and returns predefined PetNest-specific answers. This approach keeps the assistant lightweight and suitable for the academic project.

```
function getAIAnswer(question) {
    const text = question.toLowerCase();

    if (text.includes("adopt")) {
        return "Browse pets, select a pet, view details, and submit an adoption request.";
    }

    if (text.includes("volunteer")) {
        return "Visit the Volunteer section and submit the volunteer form.";
    }

    return "You can ask me about adoption, pets, volunteering, login, events, gallery, or contact.";
}
```

6.7 Implementation Steps

1. Create the HTML/CSS user interface and navigation.
2. Create MySQL tables for users, pets, adoption requests, and volunteers.
3. Configure Apache Tomcat and JSP execution.
4. Connect JSP pages to MySQL through JDBC.
5. Implement registration and login sessions.
6. Implement adoption and volunteer form processing.
7. Create admin pages for reviewing records and updating statuses.
8. Add the JavaScript-based PetNest AI Assistant.
9. Test positive, negative, and integration scenarios.

7. Testing

7.1 Manual Testing

Manual testing was performed by entering valid and invalid data into the major forms and checking the resulting page behavior and database records. The purpose was to verify that the application behaves as expected from a user's and administrator's point of view.

7.2 Test Cases

ID	Test Scenario	Expected Output	Observed Result	Status
TC001	User registration with new email	Account should be created	User record inserted	Pass
TC002	Registration with existing email	Duplicate account should be prevented	Existing email detected	Pass
TC003	Login with valid credentials	User should authenticate	Session created / success returned	Pass
TC004	Login with invalid credentials	Error should be shown	Login rejected	Pass
TC005	Submit adoption request	Request should be stored	Record inserted in adoption_requests	Pass

TC006	Admin views adoption requests	Requests should be listed	Records displayed in table	Pass
TC007	Admin approves adoption request	Status should become Approved	Status updated	Pass
TC008	Admin rejects adoption request	Status should become Rejected	Status updated	Pass
TC009	Submit volunteer application	Application should be stored	Record inserted in volunteers	Pass
TC010	Admin approves/rejects volunteer	Status should change	Status updated	Pass
TC011	Delete pet record	Selected pet should be removed	Delete action executed	Pass
TC012	AI Assistant greeting	Greeting response should appear	Response displayed	Pass
TC013	AI Assistant adoption question	Adoption guidance should appear	Relevant predefined answer	Pass
TC014	AI Assistant contact question	Contact information should appear	Configured contact response	Pass
TC015	Admin access without login	Admin page should not be freely accessible	Redirect to admin login	Pass

7.3 Testing Observations

- Form data is transferred to JSP processing pages through HTTP requests.
- PreparedStatement is used for the main database operations.
- Session attributes are used for user/admin authentication state.
- Adoption and volunteer status changes are reflected in the management pages.
- The AI Assistant works without an external service because responses are stored in JavaScript logic.

8. Enhancements

8.1 Advantages of the Project

- Simple and user-friendly navigation.
- Centralized storage of important application data.
- Online adoption-request submission reduces manual paperwork.
- Volunteer applications can be reviewed and tracked.
- Admin dashboard provides a quick summary of important counts.
- Rule-based AI Assistant provides instant answers to common questions.
- JSP and MySQL provide a straightforward academic web-development stack.
- The application can be extended with additional modules.

8.2 Limitations of the Project

- No real-time chat feature for direct communication between users and shelter admins.
- No AI-based pet recommendations, requiring users to manually search for pets.
- Lack of online payment integration for adoption fees or donations. No role-based admin access, limiting advanced permission controls.
- Mobile UI/UX improvements needed for better usability on smartphones.
- Although the system meets its primary objectives, these limitations highlight areas where enhancements can improve user experience and platform efficiency.

8.3 Future Scope

- Integrate a real-time chat system to allow direct communication between adopters and shelter admins

- Implement AI-based pet recommendations to provide personalized pet suggestions based on user preferences
- Add online payment gateways to facilitate adoption fee payments and donations securely.
- Enhance admin panel functionality by introducing role-based access control for better management.
- Develop a dedicated mobile application to improve accessibility and engagement.

By addressing these enhancements, the platform can become more interactive, efficient, and scalable, ultimately increasing adoption rates and improving the overall pet adoption process.

9. References

The project followed **IEEE/APA** citation standards and referred to various resources, including research papers, websites and tools. Some of the key references include:

Books and Research Papers:

- **Daniel Nichter**, Efficient MySQL Performance: Best Practices and Techniques
- **D. Flanagan**, JavaScript: The Definitive Guide, O'Reilly Media, 2020.
- **M. Fowler**, Patterns of Enterprise Application Architecture, Addison Wesley, 2002.

Websites & Online Documentation:

- JavaScript Documentation – <https://www.javascript.com/>
- MySQL Documentation - <https://www.mysql.com/>

